



# The interactions between genotype, management and environment in regional crop modelling



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## ARTICLE INFO

### Article history:

Received 2 January 2016

Received in revised form 10 May 2016

Accepted 10 May 2016

Available online 3 June 2016

### Keywords:

Apsim

Corn

Clustering

Spatial modelling

Uncertainty

Sensitivity

## ABSTRACT

Biophysical models to simulate crop yield are increasingly applied in regional climate impact assessments. When performing large-area simulations, there is often a paucity of data to spatially represent changes in genotype (G) and management (M) across different environments (E). The importance of this uncertainty source in simulation results is currently unclear. In this study, we used a variance-based sensitivity analysis to quantify the relative contribution of maize hybrid (i.e. G) and sowing date (i.e. M) to the variability in biomass yield ( $Y_T$ , total above-ground biomass) and harvest index (HI, fraction of grain in total yield) of irrigated silage maize, across the extent of arable lands in New Zealand (i.e. E). Using a locally calibrated crop model (APSIM-maize), 25 G × M scenarios were simulated at a 5 arc minute resolution (~5 km grid cell) using 30 years of historical weather data. Our results indicate that the impact of limited knowledge on G and M parameters depends on E and differs between model outputs. Specifically, the sensitivity of  $Y_T$  and HI to genotype and sowing date combinations showed different patterns across locations. The absolute impact of G and M factors was consistently greater in the colder southern regions of New Zealand. However, the relative share of total variability explained by each factor, the sensitivity index ( $S_i$ ), showed distinct spatial patterns for the two output variables. The  $Y_T$  was more sensitive than HI in the warmer northern regions where absolute variability was the smallest. These patterns were characterised by a systematic response of  $S_i$  to environmental drivers. For example, the sensitivity of  $Y_T$  and HI to hybrid maturity consistently increased with temperature. For the irrigated conditions assumed in our study, inter-annual weather conditions explained a higher share of total variability in the southern colder regions. Our results suggest that the development of methods and datasets to more accurately represent spatio-temporal G and M variability can reduce uncertainty in regional modelling assessments at different degrees, depending on prevailing environmental conditions and the output variable of interest.

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## 1. Introduction

Crop yield and quality are largely influenced by the interactions among genotype (G), management (M) and the growth environment (E) in agricultural systems (Hatfield and Walthall, 2015). These interactions are also represented in process-based biophysical models that are often used in climate risk assessments to simulate yield and quality of crops (Rosenzweig et al., 2013; Ewert et al., 2015). This implies that the confidence in simulation results

depends on the accurate representation of crop varieties used (i.e. G) and crop management practices (i.e. M) applied by farmers. These decisions are mostly made at the scale of individual farms and therefore can widely vary in space and time (Ewert et al., 2011). When model simulations are performed across large areas (e.g. catchments, political regions or continents), there is often a paucity of data to represent variations in G and M parameters across different E conditions (e.g. multiple locations and years). The degree by which this uncertainty in input parameters is transferred to simulated results is unknown. This is particularly important for crops such as maize (*Zea mays*) that are cultivated over a wide range of climates and a diversity of production systems worldwide (Löffler et al., 2005; Ciampitti and Vyn, 2014).

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Two key input parameters that are set at the start of maize simulations are the hybrid maturity and the sowing date (Kucharik, 2003). These G and M parameters have an important bearing on the environmental conditions to which the crop will be exposed from sowing to harvest in any given location. While the sowing date determines the start of crop development, the hybrid class maturity regulates the rate of transition between successive development stages and consequently crop growth duration. Short-season hybrids require less time, often characterised by temperature accumulation requirements (i.e. thermal-time in degree-days, °Cd), to reach reproduction and maturity. Maize growers autonomously adapt sowing dates and hybrids to prevailing environmental conditions across years and locations (Kucharik, 2003). However, this spatial and temporal variation in G and M is rarely known when performing regional maize simulations. Although the importance of this uncertainty source has been recently highlighted (Ewert et al., 2015), to the best of our knowledge, there are no studies that have systematically investigated the topic.

In this study, we use a variance-based sensitivity analysis to quantify the effects carried through to large-area maize simulation results, due to incomplete knowledge about sowing dates and hybrid maturities, across multiple locations and years. Specifically, we set up a modelling experiment to estimate the share of total variability in maize productivity and quality across the extent of New Zealand arable lands. The geographic differences in climatic suitability for maize, from nearly optimal in the north to increasingly unsuitable for maize growth in the south, creates an ideal gradient to investigate spatial patterns of response to G and M input parameters. Results of this modelling study aim to inform agricultural risk assessments that rely on limited data to characterise genotype and management of silage maize crops across large areas.

## 2. Materials and methods

### 2.1. Site and weather data

The modelling study was performed across the extent of lands classified as “suitable for arable cropping” at a 5 arc minute resolution (~5 km grid cells) in New Zealand (Fig. 1). This included ~2760 grid cells that were selected based on the Land Use Capability classification at the Land Resource Information Systems (LRIS) Portal (Newsome et al., 2008). The selected land use classes ranged from no limitation (LUC Class 1) to moderate limitation (LUC Class 4) for arable cropping. For each grid cell, 30 years of historical daily weather data (1971–2000) were used as input to run a biophysical model (Fig. 1). The Virtual Climate Station Network (VCSN) weather data was obtained from the National Institute of Water and Atmospheric Research (NIWA) of New Zealand (Sood, 2015). The daily weather input variables used in the study were air temperature (°C, maximum and minimum), total solar radiation (MJ m<sup>-2</sup>) and rainfall (mm).

### 2.2. Crop model description and testing

The release version of the maize module in the Agricultural Production Systems sIMulator (APSIM) version 7.7 (APSIM-maize) was used to perform all simulations (Holzworth et al., 2014). This modelling framework represents the response of underlying physiological processes (e.g. leaf appearance/expansion, light interception and biomass accumulation and partitioning among plant organs) of different crops to daily weather inputs and management interventions (e.g. sowing dates, fertiliser and irrigation applications). The maize model in APSIM has been previously documented ([www.apsim.info/Documentation/](http://www.apsim.info/Documentation/)) and applied across a large range of environments (Teixeira et al., 2010; Liu et al.,

**Table 1**

Parameterisation of the five hybrid types for New Zealand in APSIM-maize.

| Hybrid type     | tt.emerg.to.endjuv <sup>a</sup> (°Cd) | tt.flower.to.maturity <sup>b</sup> (°Cd) |
|-----------------|---------------------------------------|------------------------------------------|
| Very early (h1) | 130                                   | 850                                      |
| Early (h2)      | 160                                   | 875                                      |
| Mid (h3)        | 190                                   | 900                                      |
| Late (h4)       | 220                                   | 925                                      |
| Very late (h5)  | 250                                   | 950                                      |

<sup>a</sup> Thermal time from crop emergence to the end of the juvenile period.

<sup>b</sup> Thermal time from flowering to maturity. The thermal-time calculation was based on the piece-wise model by Wilson et al. (1995) used as default in APSIM-maize. The model linearly interpolates thermal-time estimates across the air temperatures of 0, 18, 26, 34 and 44°C (x-axes) with corresponding thermal-time values of 0, 10, 18, 26, 0Cd (y-axes).

2012; Archontoulis et al., 2014; Teixeira et al., 2015). In this study, the model simulations were further compared with measured data from a “sowing date by hybrid maturity” field experiment (Sorensen et al., 2000) in Hawke’s Bay, New Zealand (Fig. 1). The simulations for model testing considered management events reported in the field experiment and were performed in response to daily meteorological records measured at the experimental site. The treatments included ten different sowing dates (21 Sep to 20 Jan) and three hybrid maturities ranging from 77 to 114 units of Comparative Relative Maturity (CRM) used for classification of commercial hybrids (see Supplementary Material). The model fitness was evaluated by assessing the coefficient of determination (R<sup>2</sup>) and the root mean squared deviation (RMSD, Eq. (1)) between observed and simulated values (Kobayashi and Salam, 2000) for total silage biomass yield (Y<sub>T</sub>; i.e. above-ground dry matter containing leaves, stems, grains, rachis and husks), grain yield (Y<sub>G</sub>) and the harvest index (HI).

$$RMSD = \sqrt{\frac{1}{n} \sum_{i=1}^n (m_i - s_i)^2} \quad (1)$$

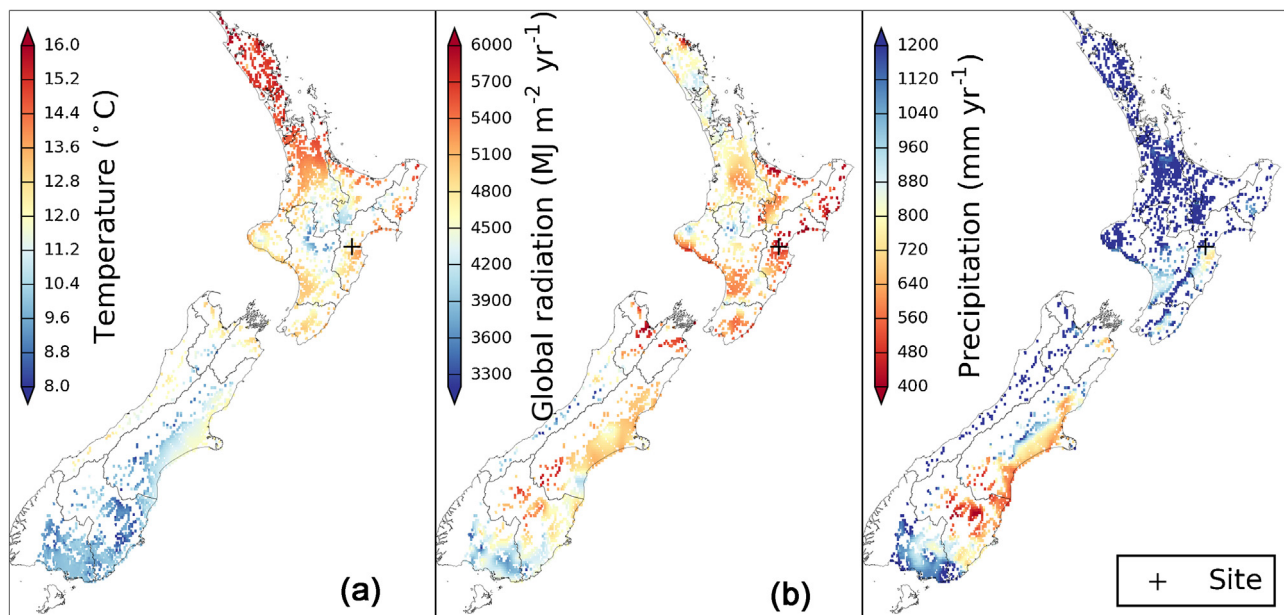
where  $n$  is the number of measurements,  $m_i$  is the measured value for observation  $i$ , and  $s_i$  is the simulated value for observation  $i$ . The results are presented as the relative RMSD (rRMSD, %) in relation to the observed mean value.

### 2.3. Simulation experiment

The simulation experiment was designed as a factorial combination of five sowing dates (1-Sep, 1-Oct, 1-Nov, 1-Dec and 1-Jan) and five hybrid maturities (Table 1). These combinations were developed to extend the sowing window slightly beyond the recommendations in seed company catalogues for New Zealand (see Supplementary Material). Recommended sowing dates range from 21 September to 3 December across most of New Zealand, with the exception of the southern regions where maize growth is constrained by low temperatures (Fig. 1). A wide range of hybrid maturities are recommended for each sowing date, depending on the climatic region. In general, very short-season hybrids are not recommended for early sowing in the northern (warmer) regions and very long-season hybrids are not recommended for the (colder) southern regions. Similarly, very early or very late sowing dates are not recommended for the most southern regions due to risk of spring and autumn frosts.

Five maturity rates were translated into two APSIM-maize parameters (“tt.emerg.to.endjuv” and “tt.flower.to.maturity”) based on the comparative relative maturity (CRM, see Supplementary Material) of New Zealand commercial hybrids (Table 1).

Two model output variables were analysed, total silage biomass yield (Y<sub>T</sub>) and the harvest index (HI). We used the HI to represent a quality aspect of maize silage because higher grain concentration



**Fig. 1.** Average climate conditions in the arable lands of New Zealand. The values of weather variables were averaged from 1971 to 2000 across 5 min grid cells. Mean daily temperature (a), annual total global solar radiation (b), and annual total precipitation (c). The site used for model testing is indicated by a cross.

increases the starch, and consequently, the metabolisable energy content of the silage (Braga et al., 2008). Therefore, higher HI potentially contributes to improved livestock biomass intake and performance due to higher forage digestibility.

A single soil type with minimum constraints to maize yields was assumed for all grid cells (Table S1 in Supplementary Material). The soil was characterised by a high water holding capacity (WHC) of 288 mm at 1.8 m depth (i.e.  $160 \text{ mm m}^{-1}$ ), with similar parameterisation across all soil layers. Organic matter content was assumed to decline from 1.8% in top layers to 0.2% at low layers, with decreasing degradability at deeper horizons. Soil water was assumed to be at field capacity at sowing. Soil conditions were re-initialised at sowing every year. Automatic irrigation and fertiliser application rules were set in the model to supply water and N in response to crop demand and minimise the degree of stress on growth rates. This reduced the influence of soil water and soil N supply on model results. By assuming irrigated conditions, we aimed simplify the analysis to assess the effects of G and M combinations on yields driven mainly by solar radiation and temperature gradients. The regional differences in rainfall regimes (Fig. 1) and access to irrigation, which largely influence maize yields and yield variability in New Zealand, are therefore beyond the scope of our analysis.

#### 2.4. Clustering of the climate conditions and sensitivity analysis

In order to study the interactions between G and M under different climate conditions, climate zones were created by *k*-means clustering (Arthur and Vassilvitskii, 2007). The clustering was applied separately to average mean daily temperature (T) and annual sum solar radiation (R). The cluster number was specified as three for both of the clustering. The clustering resulted in a layer with low temperature (LT,  $<11^\circ\text{C}$ ), median temperature (MT,  $11\text{--}13^\circ\text{C}$ ) and high temperature (HT,  $>13^\circ\text{C}$ ) (Fig. 2a) and another layer with low radiation (LR,  $<4500 \text{ MJ m}^{-2}$ ), median radiation (MR,  $4500\text{--}5000 \text{ MJ m}^{-2}$ ) and high radiation (HR,  $>5000 \text{ MJ m}^{-2}$ ) (Fig. 2b). The two layers were combined in nine climate zones: LTMR, LTMR, LTMR, MTMR, MTHR, HTLR, HTMR and HTHR (Fig. 2c).

To quantify the contribution of different factors (hybrid, sowing date and inter-annual weather conditions) to the variability of  $Y_T$  and HI, a variance-based sensitivity analysis (Santner et al., 2003)

was then applied for each grid cell. The sensitivity analysis quantified the relative share of the total variability in  $Y_T$  and HI caused by each main factor and also by the interactions among main factors. The sensitivity index ( $S_i$ ) was calculated as the sums of squares for each factor in relation to the total sums of squares from ANOVA decompositions (Welch et al., 1992; Sobol', 1993; Santner et al., 2003). Statistical analyses were performed using the software R (Core Team, 2012). For box-and-whisker type plots, the upper and lower box edges represent the first and third quartiles (the 25th and 75th percentiles). Whiskers represent the 5th and 95th percentiles and data beyond the end of the whiskers are identified as outliers.

### 3. Results

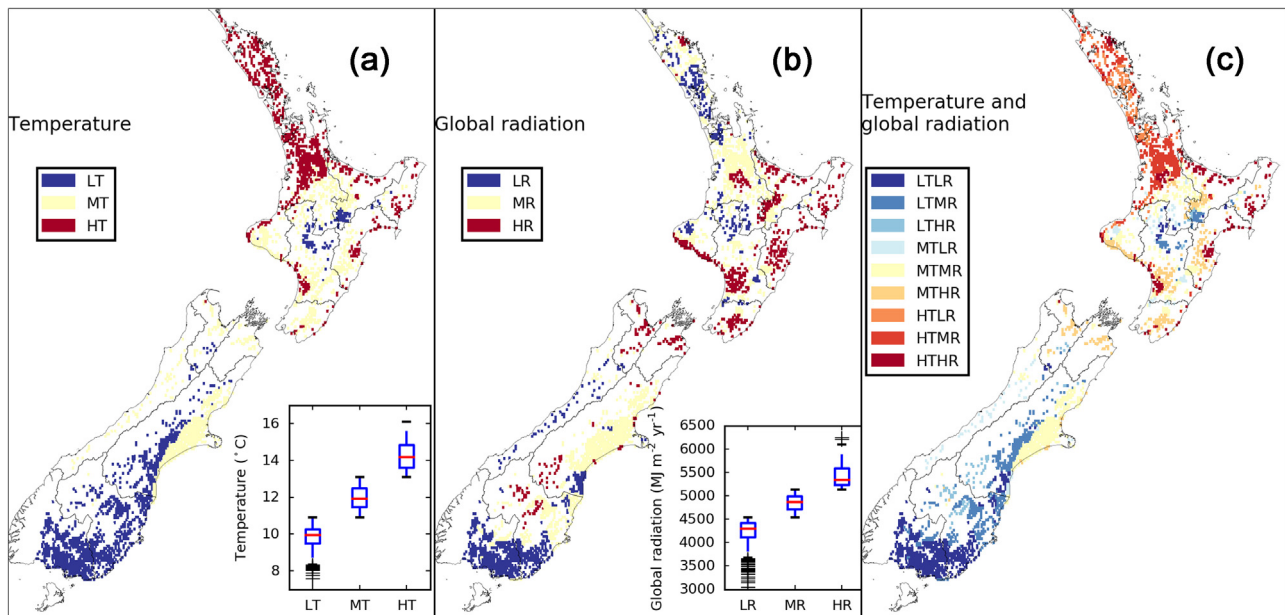
#### 3.1. Model testing

When compared to field measurements, APSIM-maize simulations followed similar magnitude and direction of change in response to the multiple sowing date and hybrid maturity treatments, over the wide range of measured values of  $Y_T$  ( $10\text{--}35 \text{ t ha}^{-1}$ ),  $Y_G$  ( $1.5\text{--}18 \text{ t ha}^{-1}$ ) and HI ( $0.3\text{--}0.58$ ) (Fig. 3). The  $R^2$  and the rRMSD between simulated and observed values were, respectively, 0.7 and 18% of the mean for  $Y_T$ , 0.7 and 26% for  $Y_G$  and 0.9 and 22% for HI.

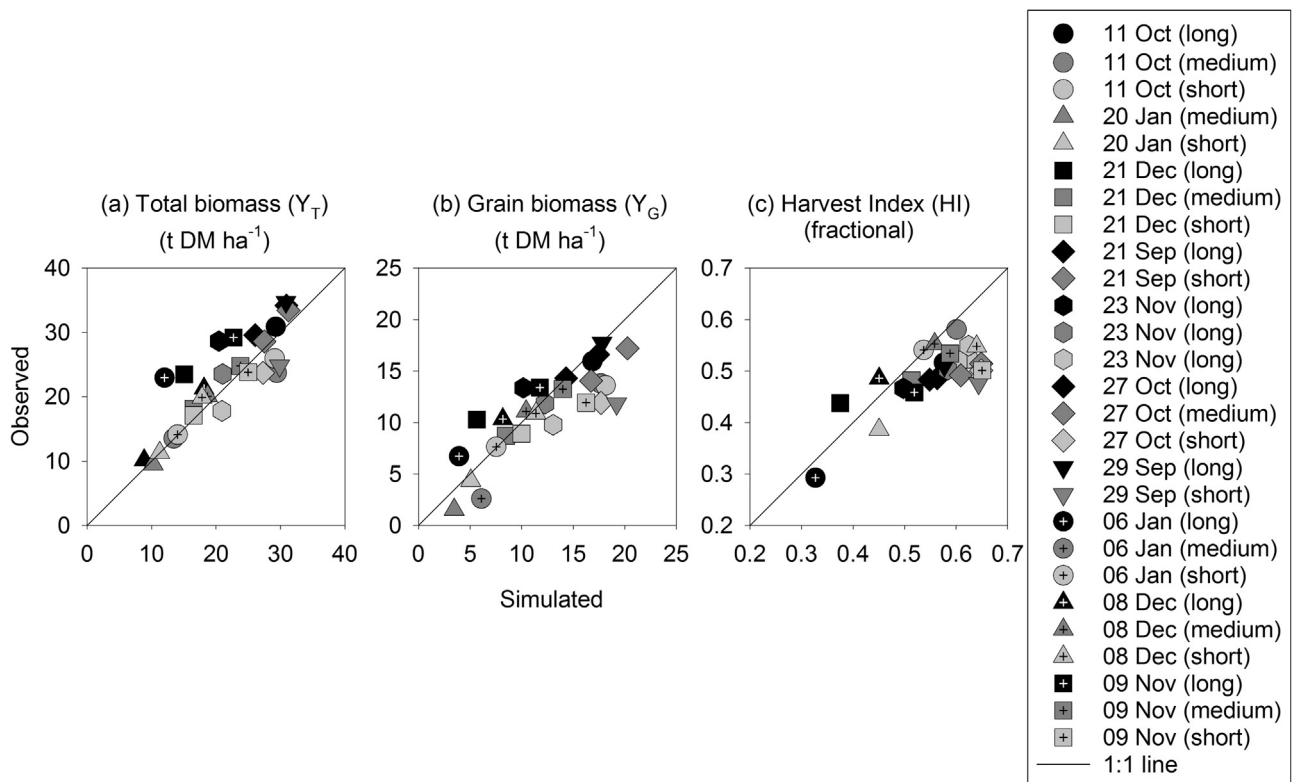
#### 3.2. Spatial patterns of means and variability for yield and harvest index

Pooled across the 25 G x M combinations for each grid cell, the highest silage biomass yields ( $>20 \text{ t ha}^{-1}$ ) were simulated in the mid-latitudes, intermediary yields ( $17\text{--}19 \text{ t ha}^{-1}$ ) in the upper north and the lowest yields ( $<16 \text{ t ha}^{-1}$ ) in the south (Fig. 4). The HI was highest ( $>0.50$ ) in the northern/central regions and lowest ( $<0.35$ ) in the southern regions. The variability (CV) across the 25 G x M scenarios, for both variables, increased from north to south (Fig. 4).

Average values within individual climate zones ranged from 11 to  $23 \text{ t ha}^{-1}$  for  $Y_T$  and from 0.23 to 0.50 for HI (Table 2). Mean HI consistently increased from low (L) to high (H) temperature climate zones. The mean  $Y_T$  responded positively to the increase in both radiation (R) and temperature (T). The variability (CV) was reduced



**Fig. 2.** Climatic zoning for irrigated crop conditions across New Zealand arable lands. The three temperature (a) and three global solar radiation (b) clusters were overlapped to generate nine climate zones combinations (c).



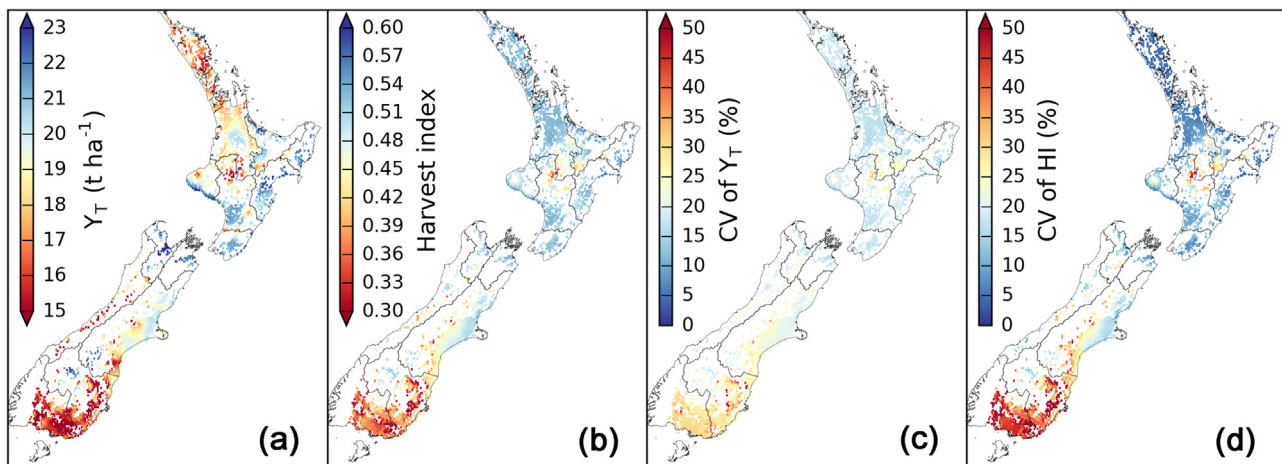
**Fig. 3.** Comparison of simulated and observed (a) silage maize biomass (b) grain biomass and (c) harvest index for 5 different sowing dates and 3 hybrid maturities in Hawke's Bay, Napier, New Zealand.

from 44 to 20% for  $Y_T$  and from 79 to 15% for HI as temperature and radiation increased from LTLR to HTHR climate zones.

The  $Y_T$  and HI showed contrasting patterns of response to G x M combinations, depending on the climate zone (Fig. 5). For example, for early sowing (s1) in LTLR climate zones (Fig. 5a),  $Y_T$  increased from 10 to 26 t ha<sup>-1</sup> when hybrids changed from short- (h1) to long-maturity (h5). In contrast, for late sowing date (s5), longer season hybrids had negligible impact on  $Y_T$  (e.g. ~10 t ha<sup>-1</sup>).

The same analysis for the HTHR climate zone showed that  $Y_T$  was responsive to both early (~22–26 t ha<sup>-1</sup>) and late sowing dates (~14–18 t ha<sup>-1</sup>). Within each temperature class, the mean  $Y_T$  always increased from low to high radiation. In the HTHR climate zone, highest  $Y_T$  values were estimated for early sowing (s1) and long season (h5) hybrids. In contrast, for the LTLR climate zone highest  $Y_T$  occurred with intermediary sowing dates (s2 and s3) and hybrid maturities (h2 to h4). The HI was less responsive to G





**Fig. 4.** Maps of simulated mean total aboveground biomass,  $Y_T$  (a), mean harvest index, HI (b), coefficient of variation (CV) of  $Y_T$  (c) and CV of HI (d) for 30 years and 25 G x M scenarios. The CV calculation used mean values across years, so that it represents variability from G x M scenarios.

**Table 2**  
Pooled responses of silage maize yield ( $Y_T$ ) and harvest index (HI) to climate conditions of temperature (T) and radiation (R) in nine climate zones. Number of grid cells in each climate zone was determined by k-means clustering. The medians, percentile ranges (25th and 75th percentile values inside parenthesis) and the coefficient of variation (CV) were calculated from 30 year averages for each of the 25 scenarios (5 hybrid maturities and 5 sowing dates) within each grid cell.

| Climate Zone | Grid cell number<br>(n) | Averages <sup>a</sup>  |                            | Coefficient of variation (CV) |                   |
|--------------|-------------------------|------------------------|----------------------------|-------------------------------|-------------------|
|              |                         | Yield ( $t\ ha^{-1}$ ) | Harvest Index (fractional) | Yield (%)                     | Harvest Index (%) |
| LTLR         | 589                     | 11(8–15)               | 0.23(0.05–0.37)            | 44(40–44)                     | 79(68–83)         |
| LTMR         | 480                     | 13(9–17)               | 0.25(0.11–0.37)            | 45(36–52)                     | 73(55–86)         |
| LTHR         | 95                      | 15(10–20)              | 0.28(0.17–0.41)            | 42(34–49)                     | 63(50–72)         |
| MTLR         | 122                     | 17(14–20)              | 0.39(0.32–0.5)             | 30(27–33)                     | 40(32–49)         |
| MTMR         | 533                     | 19(16–23)              | 0.42(0.35–0.52)            | 29(27–31)                     | 37(33–41)         |
| MTHR         | 354                     | 22(19–26)              | 0.44(0.42–0.52)            | 26(24–28)                     | 29(25–33)         |
| HTLR         | 132                     | 19(17–21)              | 0.50(0.49–0.54)            | 21(20–21)                     | 13(11–14)         |
| HTMR         | 305                     | 20(18–23)              | 0.50(0.49–0.54)            | 21(20–21)                     | 14(12–17)         |
| HTHR         | 151                     | 23(21–26)              | 0.50(0.49–0.54)            | 20(19–22)                     | 15(12–19)         |

<sup>a</sup> Values are calculated from 30 year means for each grid cell considering 25 hybrid by sowing date scenarios within nine climate zones classified as high (H) or low (L) temperature (T) and solar radiation (R) across New Zealand arable lands.

and M combinations than  $Y_T$  in warmer climate zones but showed a strong response in LT climate zones, particularly for later sowing dates.

### 3.3. Sensitivity of $Y_T$ and HI to G, M and e

The spatial distributions of the sensitivity index ( $S_i$ ) for  $Y_T$  and HI are shown in Fig. 6. The  $Y_T$  was consistently more sensitive to sowing date (M) in the central and northern areas of the country. In contrast, the response of HI to sowing dates was the lowest in the northern regions as further illustrated by the pooled  $S_i$  values in HT zones (Fig. 7). The sensitivity of both variables to hybrid maturity was higher in northern warmer regions (Fig. 6) as shown by the increase of  $S_i$  in HT climate zones (Fig. 7).

The sensitivity to inter-annual weather variability showed a less variable pattern across the country. In contrast, the sensitivity to the interactions among factors was spatially heterogeneous and also differed between  $Y_T$  and HI. Interactions were more important in the southern regions for  $Y_T$  and in the northern regions for HI (Fig. 6).

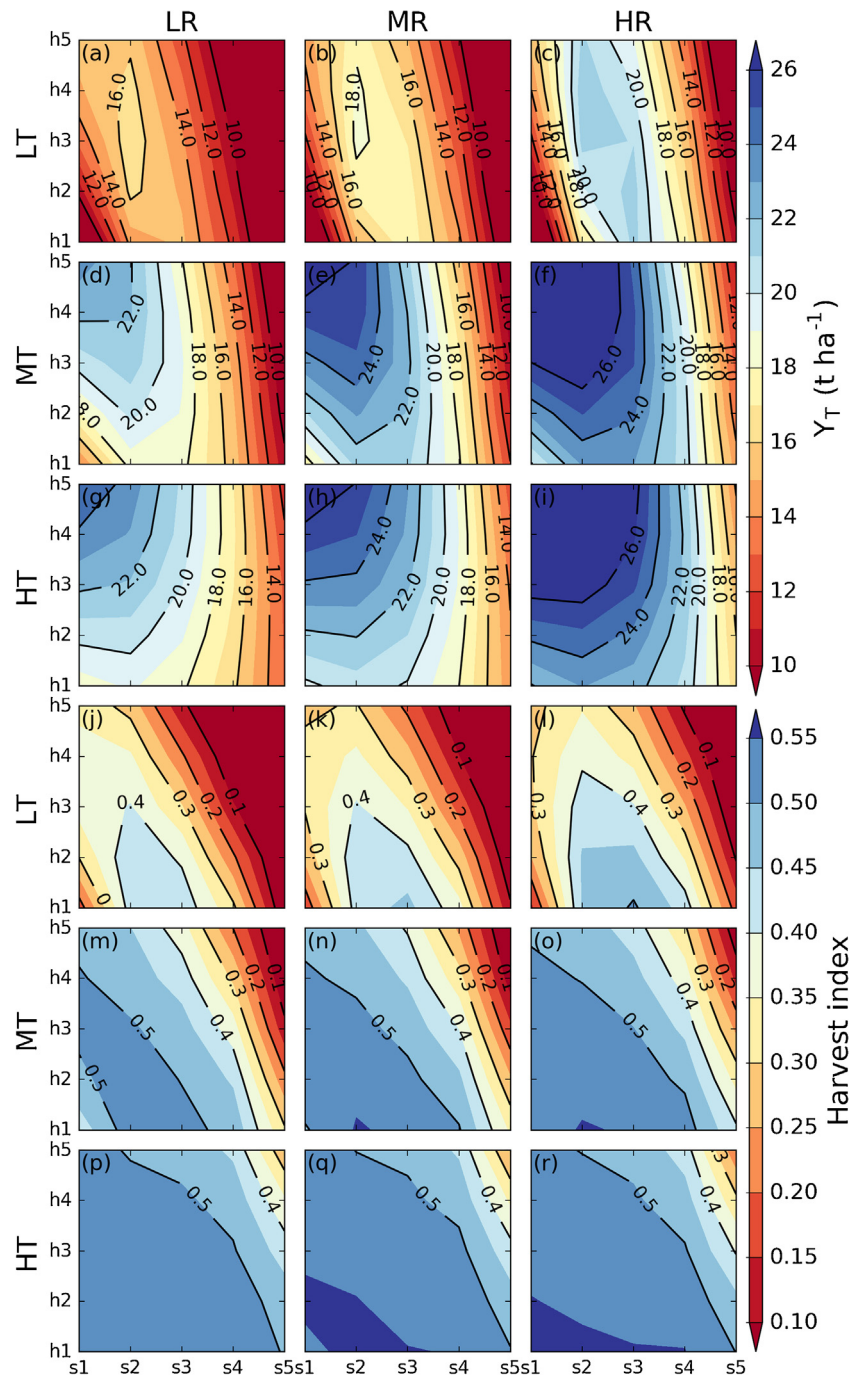
## 4. Discussion

The understanding of uncertainty sources that influence crop yield simulations is critical to improve agricultural risk assessments (Walker et al., 2003; Bennett et al., 2013). In this case study, we quantified the main effects and interactions of genotype (G) and management (M) input parameters on crop model outputs across

contrasting environments (E) in New Zealand. To the best of our knowledge, this is the first study in which such systematic analysis is performed.

As a prerequisite for performing the proposed simulation experiment, the responsiveness of simulated  $Y_T$  and HI was tested against measured field data from a hybrid maturity by sowing date experiment (Fig. 3). When compared to a recent broad evaluation of APSIM-maize in North America (Archontoulis et al., 2014), our results showed a similar accuracy for  $Y_T$  estimation (rRMSD = 18%) but lower for  $Y_G$ , and consequently for HI (rRMSD = 22%), due to an overestimation at the highest values ( $Y_G > 12\ t\ ha^{-1}$ ). The model responded to multiple sowing dates and hybrid maturities across a sufficiently wide range of measured  $Y_T$  ( $10\text{--}35\ t\ ha^{-1}$ ) and HI ( $0.3\text{--}0.6$ ). The model's fitness-for-purpose for this simulation study was further indicated by the sensible spatial patterns of maize productivity (Fig. 4) and the magnitude and direction of responses to G and M combinations (Fig. 5) that align with experimental observations of maize growth in New Zealand (Wilson et al., 1994; Tsimba et al., 2013a; Tsimba et al., 2013b).

A first key insight from our analysis is that the sensitivity of model outputs to G and M parameters differs across E conditions. This was shown by the systematic changes in  $Y_T$  and HI response to hybrids and sowing dates across climate zones (Fig. 5). These results highlight the importance of considering local environmental conditions when setting input parameters in crop models (Angulo et al., 2013). For the near-optimal supply of water and N assumed in our simulations, solar radiation is the main determinant of biomass accumulation, while temperature regulates light interception (via



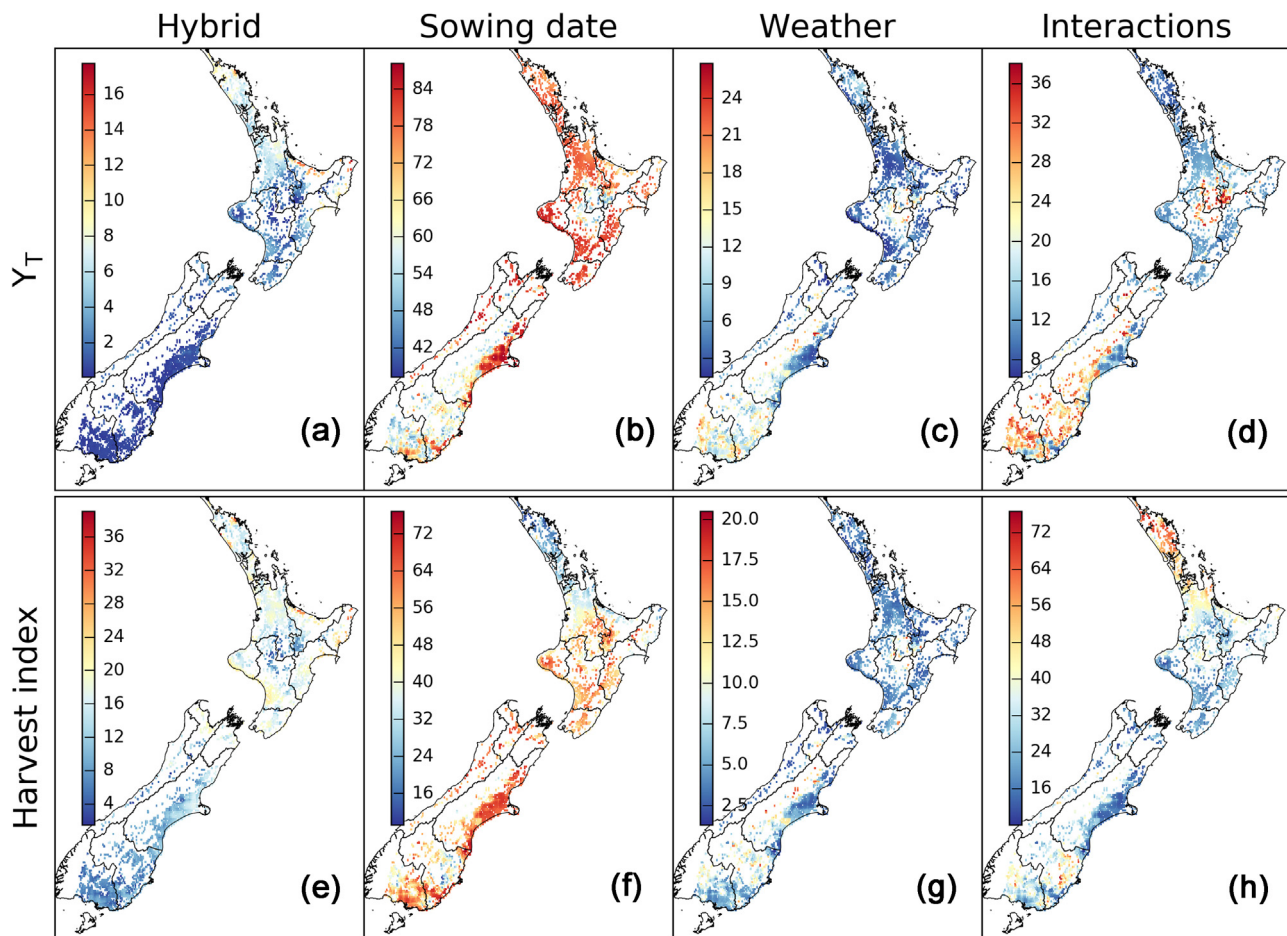
**Fig. 5.** The interaction effects of delaying sowing date (s1 to s5) and increasing the hybrid thermal-time requirement for flowering (h1 to h5) on total aboveground biomass of silage maize ( $Y_T$ ; panels a–i) and harvest index (HI; panels j–r) across nine climate zones with low (L), median (M) or high (H) combinations of temperature (T) and radiation (R).

canopy development rates) and the efficiency of light conversion into biomass (Monteith, 1972; Wilson et al., 1995; Teixeira et al., 2014). This explains the relatively higher sensitivity of  $Y_T$  to sowing dates under the more favourable growth conditions in the central and northern warmer regions. In contrast, HI was more sensitive to sowing dates at higher latitudes in the south. This occurred because early-spring and late-summer sowing dates exposed crops to low temperatures and risks of frost, particularly in the LT climate zones, more frequently inducing crop failure or compromising grain formation. Such systematic patterns of response to environmental conditions can be further exemplified by the relationship between

the sensitivity index ( $S_i$ ) and the average annual temperature in each grid cell (Fig. 8).

There were clear systematic patterns of response of  $S_i$  values to annual average air temperature for both variables (Fig. 8). For example, the sensitivity to hybrid maturity increased from negligible values at 8°C to ~15% for  $Y_T$  and ~20% for HI at 16°C. These curvilinear patterns of response, which are specific to the conditions in our case-study, largely differed among the input factors and showed considerable variability for two output variables.

The second key insight gained from our results is that the sensitivity to G and M parameterisation differs between output variables (Fig. 5). The contrasting sensitivity between  $Y_T$  and HI, depending



**Fig. 6.** Spatial pattern of the sensitivity index ( $S_i$ ) for total above ground biomass ( $Y_T$ ; a–d) and harvest index (e–h) in response to hybrid maturities (a, e), sowing dates (b, f), inter-annual variability (c, g) and interactions (d, h) across arable lands in New Zealand.

on climate zone, is a reflection of the non-linear representation of physiological processes in the crop model. Similarly, for crop rotation simulations in New Zealand, the sensitivity of model results to input parameters was found to differ with the model output of interest (Teixeira et al., 2015; Teixeira et al., 2016). The more stable nature of HI in comparison with  $Y_T$  in HT climate zones (Fig. 5) agrees with field observations in temperate regions of Argentina and New Zealand (Otegui et al., 1995; Wilson et al., 1995). These studies showed low variability in HI across a range of sowing dates and hybrid maturities, as long as light interception or light use efficiency during the grain filling period were not drastically affected.

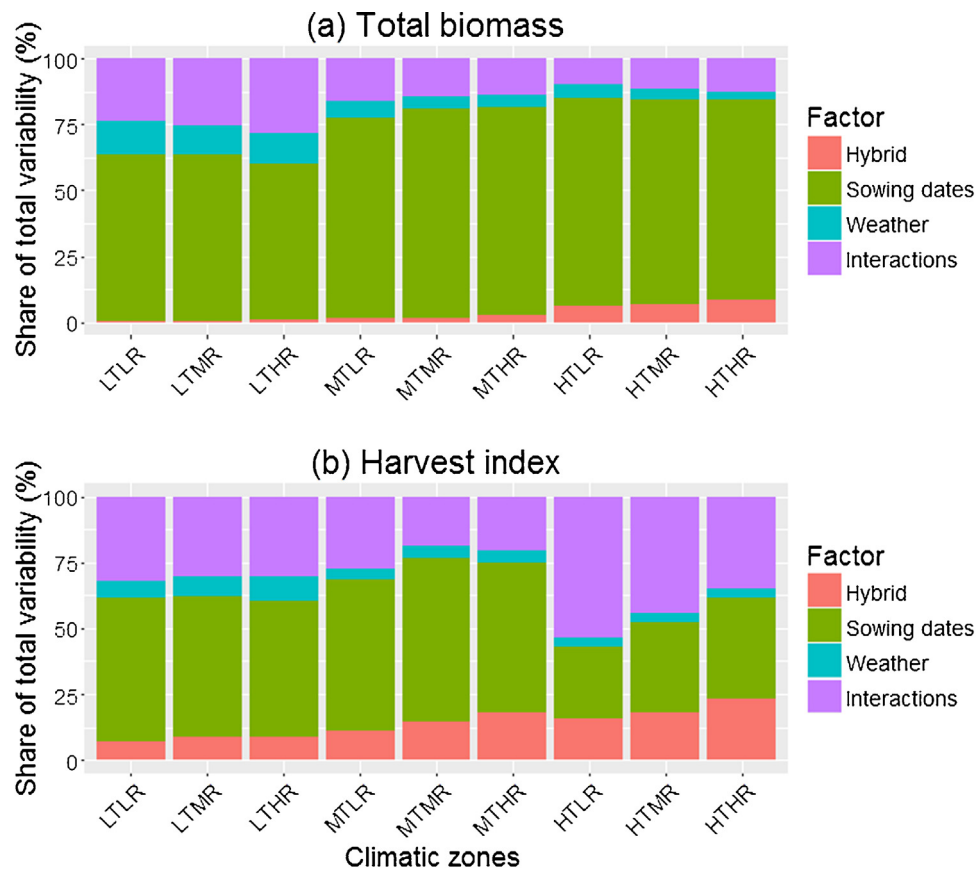
An implication of our findings is that the degree by which a more accurate spatial and temporal representation of G and M parameters reduces uncertainty in regional risk assessments depends on E conditions. Different approaches have been used to deal with the paucity of spatial data on G and M parameters in previous studies, for example, by assuming constant parameters across a region (Zhao et al., 2015), scaling survey data (Trnka et al., 2014), considering multiple parameter combinations (Kucharik, 2003) or using environmental and crop proxies to estimate input parameters (Kucharik, 2003; Waha et al., 2012; Teixeira et al., 2013). Therefore, efforts to develop more accurate datasets (e.g. Braga et al., 2008; Portmann et al., 2010) and dynamic methods to simulate G and M parameters (e.g. Teixeira et al., 2011; Waha et al., 2012) are two main routes to reduce simulation uncertainty. Future approaches may consider decision rules that make hybrid choice also a function of estimated sowing dates to capture the logic used by maize growers to minimise production risks. Of note is that

part of the uncertainty in G and M combinations is due to socio-economic drivers of decision making (e.g. farmer's preferences, machine availability, market signals and regional policy regulations) which cannot be derived from simple rules in response to environmental conditions. This reinforces the value of exploring a wide range of sensible G and M combinations, with or without the attribution of likelihood (e.g. Kucharik, 2003), given the complex nature of adaptation by farmers at the local scale (Patt et al., 2010). It is also worth noting that farmers will continuously adapt to changes in climatic conditions by altering crop management, including sowing dates and genotypes. Thus impact assessments of future climate should consider making these not only spatially but also temporally variable in response to prevailing environmental conditions.

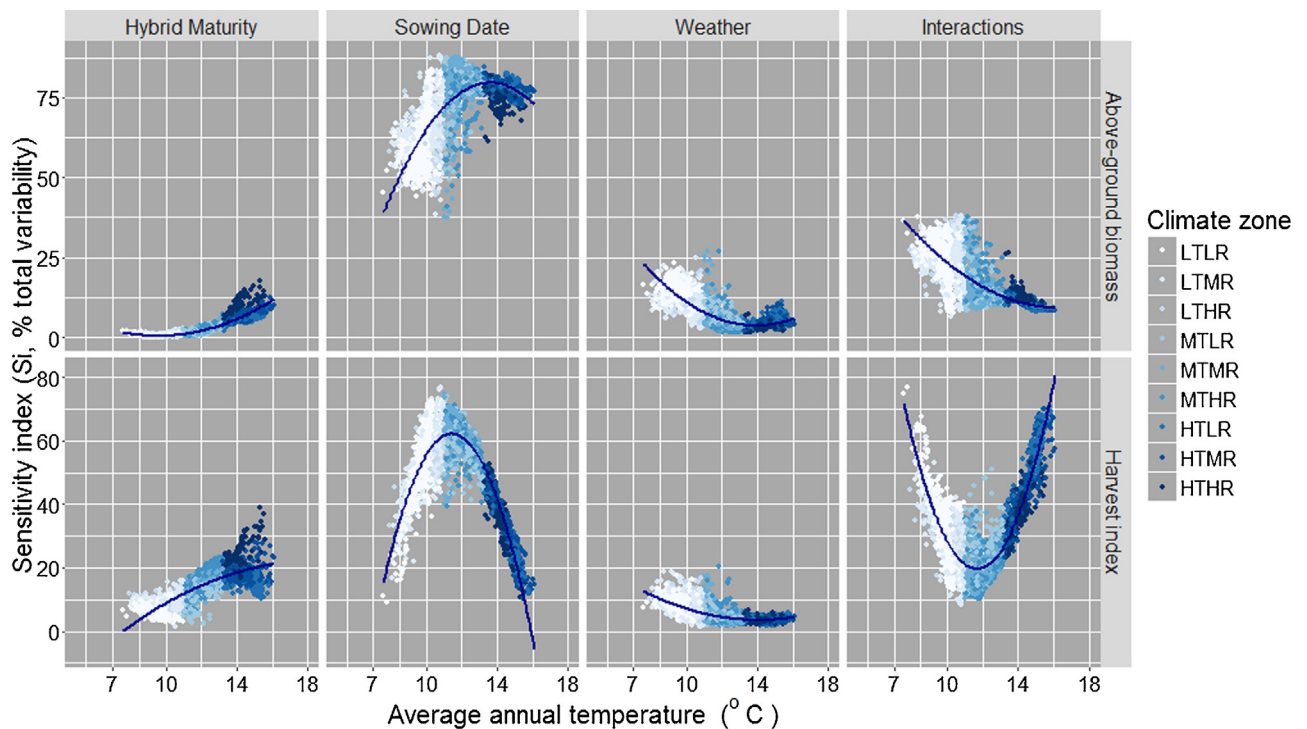
## 5. Conclusions

Our results indicate that the sensitivity of simulation results to genotype (G) and management (M) responds systematically to environmental conditions (E). Limited knowledge on the variability of G and M was particularly important under marginal growth conditions, such as the colder southern regions of New Zealand. The sensitivity of model outputs to G and M parameters also differed between output variables. Therefore, when performing sensitivity analysis, it is important to consider the response of individual output variables under the range of environmental conditions present in the simulation study. These results also emphasise the value of developing improved datasets and method-





**Fig. 7.** The sensitivity of total aboveground biomass ( $Y_T$ , a) and harvest index (HI, b) to hybrid maturities, sowing dates, inter-annual weather variability and interactions. Values were pooled for climate zones considering a combination of high (H) and low (L) temperature (T) and radiation (R) across New Zealand arable lands.



**Fig. 8.** The sensitivity of total above-ground biomass yield ( $Y_T$ ) and harvest index (HI) to hybrid maturity, sowing date, inter-annual weather variability, and the interactions among these factors, in response to the average annual temperature in each 5 km grid cell across climate zones across New Zealand arable lands. Climate zone classification considers a combination of high (H) and low (L) temperature (T) and radiation (R) within each individual grid cell. Lines illustrate empirical response patterns of  $S_i$  to air temperature.



ologies to more accurately represent variability in genotype and management across locations and time periods. In conjunction, these efforts are expected to reduce the uncertainty in large-area agricultural assessments that require data from biophysical models.

## Acknowledgements

This work was supported by funding from the Climate Change Impacts and Implications for New Zealand project (CCII) financed by the Ministry of Business, Innovation and Employment (MBIE) and the Royal Society of New Zealand. The authors thank Mr. Allister Holmes (FAR, Bay of Plenty) for technical evaluation of model results, Dr. Heidi Webber (INRES-Bonn) for technical discussions on the manuscript concepts, Dr. Joanna Sharp (PFR-NZ) for assistance with soil parameterisation and Ms. Helen Hulme (PFR-NZ) for support with maize validation datasets. E. Teixeira thanks the LAP-INRES group (Bonn University) for hosting the collaboration visit for the development of this study. GZ was supported by the German Federal Ministry of Education and Research (BMBF) through the SPACES project “Living Landscapes Limpopo” and WASCAL (West African Science Service Center on Climate Change and Adapted Land Use) project.

## Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.eja.2016.05.005>.

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